

1. Consider the clarity of the following proposition (Is it nice to read? What would you change about how its written while maintaining the same truth value?), then express this proposition as a sentence using symbols from propositional logic such as the quantifiers $\forall$ and $\exists$ or the connectives $\wedge, \vee, \neg$.

"If every natural number less than 50 has at most three distinct prime factors and the largest prime number smaller than 50 is 47, then the sum of the distinct primes dividing n is less than 141 for every n less than 50."

Also find the negation of this proposition both as formal logical symbols and as an English sentence.

Solution: We can express this proposition as

$$(\forall(n \in \mathbb{N}, n < 50) \neg (\exists(p_1, p_2, p_3, p_4 \text{ unique primes})(p_i \text{ divides } n \text{ for each } i)) \wedge (\forall(p \text{ prime}, p < 40)(p \leq 47))) \\ \rightarrow \forall(n \in \mathbb{N}, n < 50)(\text{The sum of the distinct primes dividing } n < 141)$$

Or any other logically equivalent proposition. The negation can be found as the negation of an implication $p \rightarrow q$, i.e. by $p \wedge \neg q$.

2. Suppose that we have 4 propositions p, q, r, s for which we know the following compound propositions are true

$$p \rightarrow q, \quad q \rightarrow r, \quad r \rightarrow s, \quad \text{and} \quad s \rightarrow p.$$

Show that all of the propositions p, q, r, s are logically equivalent to each other.

Solution: Let us consider showing $p \leftrightarrow q$. We already have $p \rightarrow q$ so we need only to show that $q \rightarrow p$. Note that $q \rightarrow r$ and $r \rightarrow s$ implies that $q \rightarrow s$ and together with $s \rightarrow p$ we find that $q \rightarrow p$ as desired.

3. Prove that every integer n is either a multiple of 3, one more than a multiple of 3, or 2 more than a multiple of 3. Using this, proceed to prove that for every integer n , we have that $n^4 - n^2$ is a multiple of 3.

Solution: Suppose that n is not a multiple of 3 and $n - 1$ is also not a multiple of 3. Then $n - 2, n - 1, n$ forms a string of three consecutive integers and so one of them must be multiple of 3, hence we have that $n - 2$ is a multiple of 3. Thus we see that either n is a multiple of 3 or $n - 1$ is a multiple of 3 or $n - 2$ is a multiple of 3.

Now we have that $n = 3k$ or $n = 3k + 1$ or $n = 3k + 2$ for some integer k . In each case we can respectively show that $n^4 - n^2 = 3(\text{some integer})$.

4. Define the average of a set of k different integers $\{n_1, \dots, n_k\}$ to be their sum divided by k , i.e. the formula $(n_1 + \dots + n_k)/k$. Prove or disprove that if a collection of k integers $\{n_1, \dots, n_k\}$ has an average of N where N is not an integer, there exists an integer n_i in this collection such that $n_i < N$.

Solution: Suppose this was not the case and we had a collection of integers $\{n_1, \dots, n_k\}$ with an average N which is not an integer and every n_i in this set satisfied $n_i \geq N$. Then since N is not an integer we must have that $n_i > N$ for each i and we have that $(n_1 + \dots + n_k) > N + \dots + N = k \cdot N$. It follows that our set has an average which is strictly greater than N , a contradiction.